

Theoretical Remarks on Feudal Hierarchies and Reinforcement Learning

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Reinforcement Learning

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Reinforcement Learning

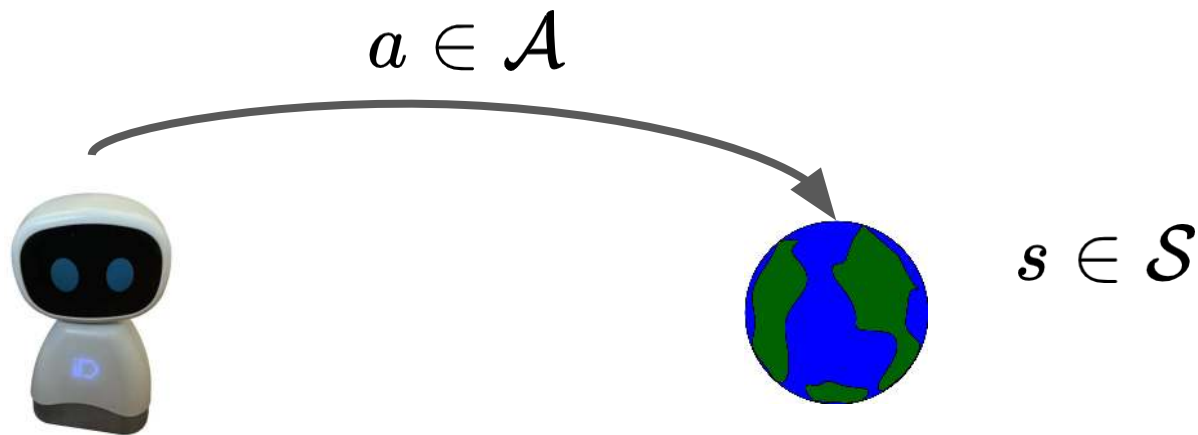


Reinforcement Learning

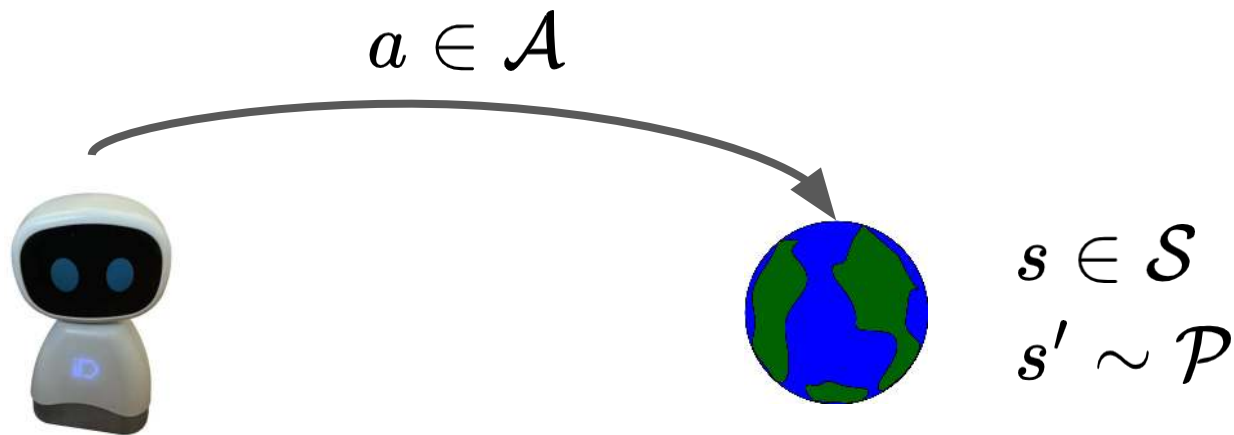


$$s \in \mathcal{S}$$

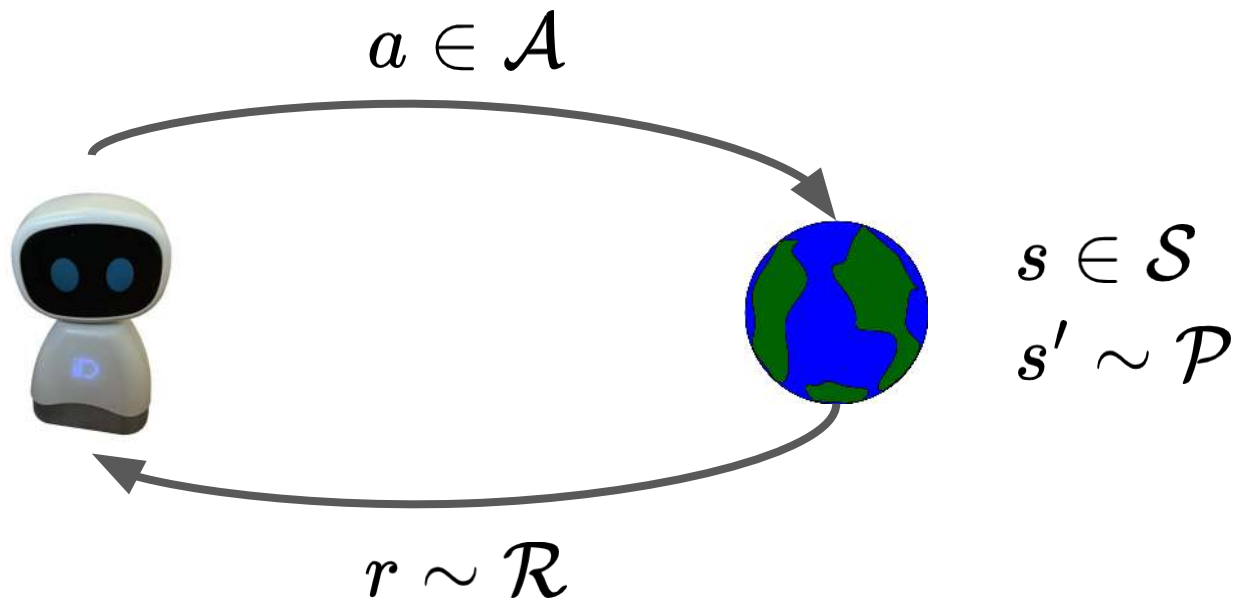
Reinforcement Learning



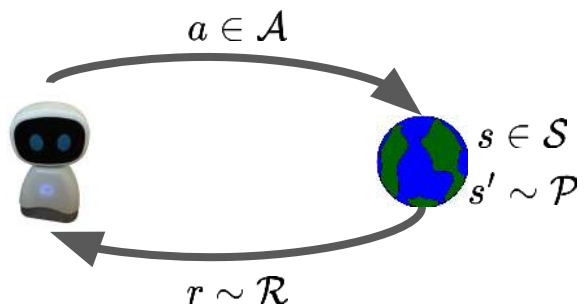
Reinforcement Learning



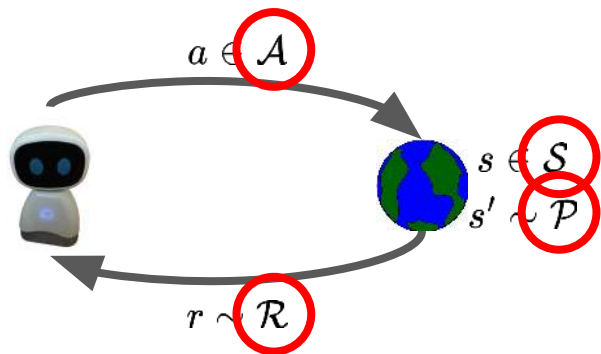
Reinforcement Learning



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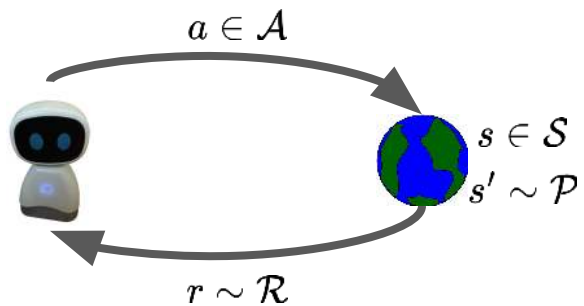
Reinforcement Learning



MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$$

Reinforcement Learning



MDP

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Policy

$$\pi : \mathcal{S} \rightarrow \mathcal{A}$$

Reinforcement Learning

**Value-
function**

$$q^{\pi}(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

Reinforcement Learning

**Value-
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$$q^{\pi}(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

Solution

$$q^* : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$$

Reinforcement Learning

**Fixed-point
equation**

$$q^*(s, a) = \mathbb{E} [r(s, a) + \gamma \max_{a'} q^*(s', a')]$$

Reinforcement Learning

**Fixed-point
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$$q^*(s, a) = \mathbb{E} [r(s, a) + \gamma \max_{a'} q^*(s', a')]$$

Q-learning

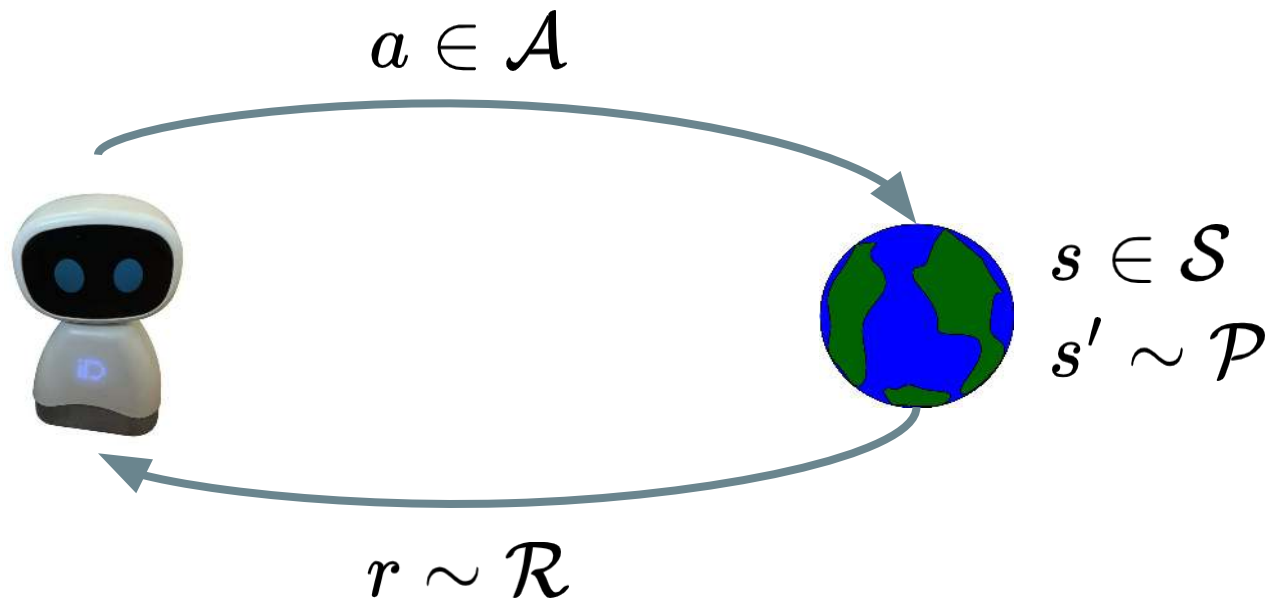
$$q_{t+1}(s_t, a_t) = (1 - \alpha) q_t(s_t, a_t) + \alpha (r_t + \gamma \max_{a'} q_t(s'_t, a'))$$

Convergence

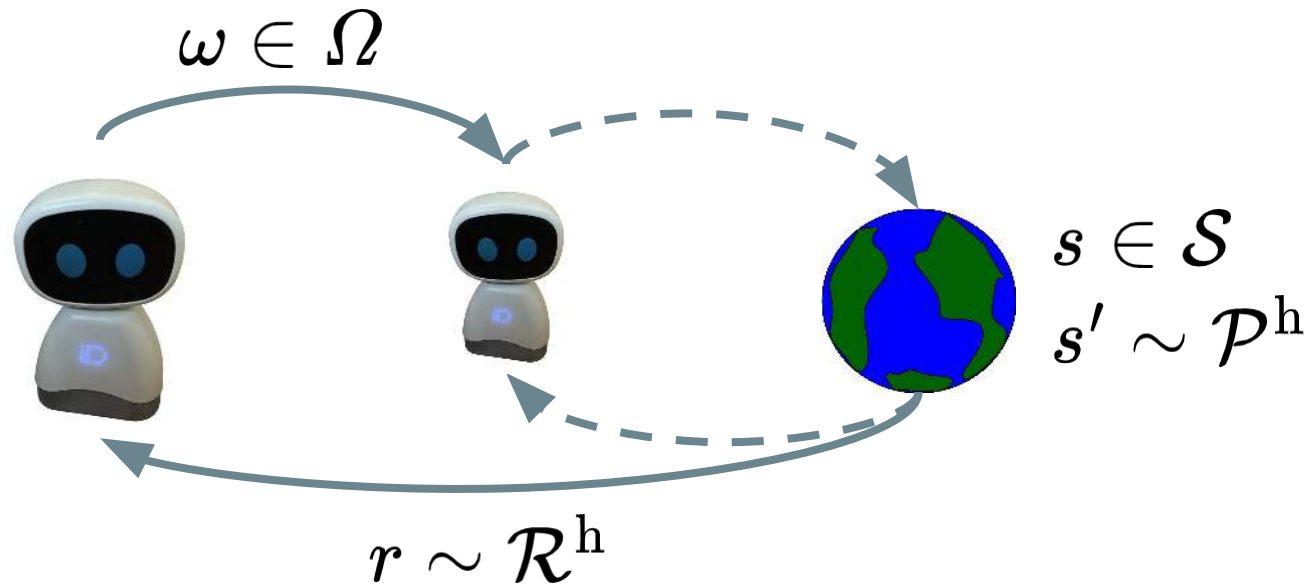
Q-learning converges to the optimal solution.

Feudal Hierarchies

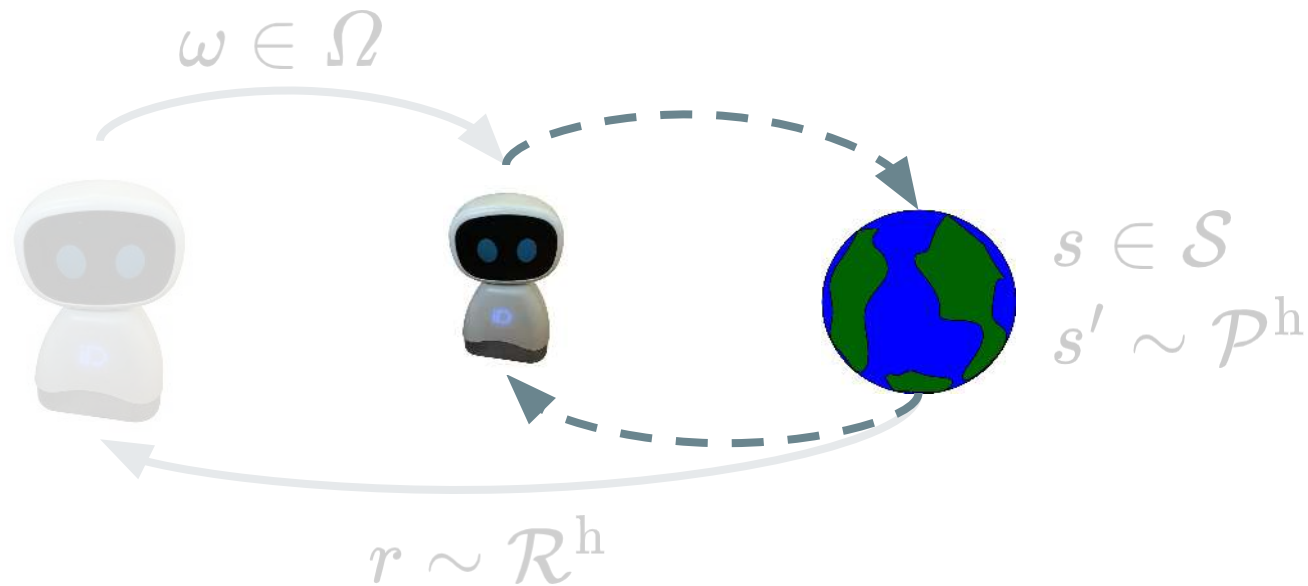
Feudal Hierarchies



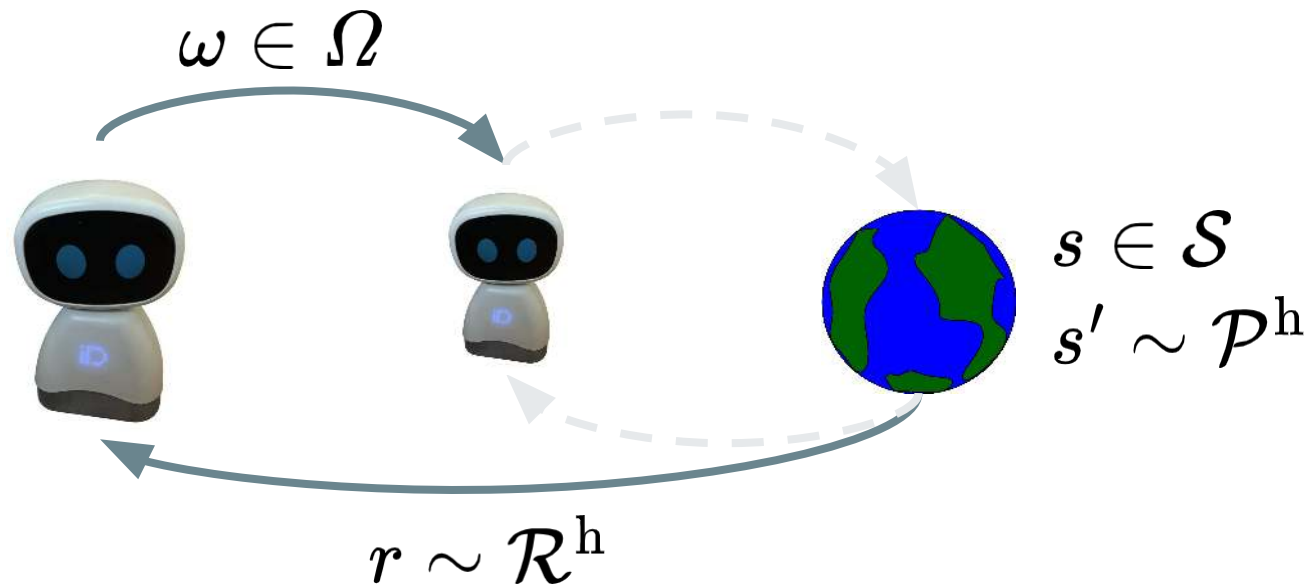
Feudal Hierarchies



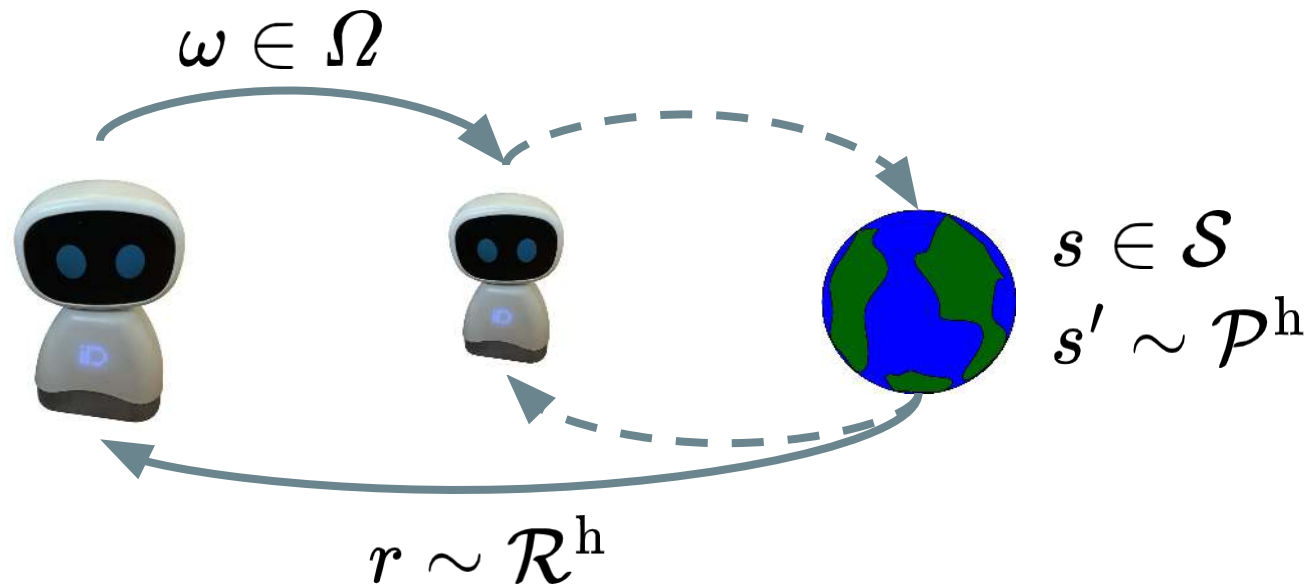
Feudal Hierarchies



Feudal Hierarchies



Feudal Hierarchies



Non-stationarity

The MDP depends on the low-level policy.

Research Question

Does Q-learning solve feudal hierarchies?

Remark 1

The MDP converges if the policy converges.

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proof.

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We show that each component of the MDP is a continuous function of the policy.

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Equivalently, we conclude the MDP converges as the policy converges.

Remark 2

Q-learning converges in convergent MDPs.

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proof:

Q-learning errors have the form $\Delta_{t+1} = (1 - \alpha) \Delta_t + \alpha F_t$
where $\Delta_t = q_t - q^*$
and $F_t = r_t + \gamma \max_{a'} q'_t - q^*$

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and $F_t = r_t + \gamma \max_{a'} q'_t - q^*$

We show that $\|F_t\| \leq \gamma \|\Delta_t\| + \epsilon_t, \epsilon_t \rightarrow 0$

Therefore, we conclude that $\Delta_t \rightarrow 0$

Conclusion

Q-learning solves feudal hierarchies.

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